

Harsha Kanaparthi

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Software engineer with 2.5+ years of industry experience building scalable distributed systems, full-stack applications, and AI-powered features in Python, Java, Go, TypeScript, and React. Shipped production systems processing 10K+ events/sec at sub-100ms latency on AWS with Docker and CI/CD. IEEE-published researcher. M.S. Computer Science, GPA 4.0.

EDUCATION

University of North Carolina at Charlotte <i>M.S. Computer Science (Data Science & InfoSec); GPA: 4.0/4.0</i>	Charlotte, NC Aug. 2024 – May 2026
Koneru Lakshmaiah University <i>B.S. Computer Science and Engineering; GPA: 9.47/10.0</i>	Vijayawada, India Aug. 2020 – May 2024

EXPERIENCE

Graduate Teaching Assistant <i>University of North Carolina at Charlotte</i>	Aug. 2025 – May 2026 Charlotte, NC
- Supported 110+ students in Survey of Programming Languages (Go, Haskell, Swift) and Software Systems Design & Implementation; A-grade rate improved from 70% to 90% through structured mentoring and code review - Designed assignments with 48-hour grading turnaround; collaborated cross-functionally with faculty to deliver course solutions and maintain platform reliability across two semesters	
Machine Learning Engineer <i>Lineysha & Thevan Software Company</i>	Jun. 2023 – Dec. 2023 Andhra Pradesh, India
- Built full-stack production system with Python Flask REST API and React/TypeScript frontend; designed cosine similarity search across 10K+ face embeddings achieving 99.4% accuracy at <500ms latency with MySQL and Redis - Reduced inference time 40% through batch processing, embedding caching, and optimized database indexing; implemented JWT authentication and role-based access control serving concurrent users at 99%+ reliability - Deployed VGG-Face CNN with transfer learning for real-time facial recognition; wrote comprehensive unit tests and API documentation ensuring production-grade code quality	
Software Engineer <i>LogicMind</i>	Jan. 2021 – Jun. 2023
- Designed and shipped scalable REST APIs and backend services in Python, Go (Gin), and Java supporting 3K+ concurrent users; owned AWS (EC2, S3, RDS) deployments with Docker and GitHub Actions CI/CD achieving 99.5% uptime - Reduced API latency 25% through PostgreSQL query optimization, connection pooling, and Redis caching; built event-driven analytics pipelines processing 100K+ gameplay events daily for real-time insights - Implemented AI-driven matchmaking algorithm improving average session duration by 14%; collaborated cross-functionally in Agile sprints to design, test, and ship production features end-to-end	

PROJECTS

Stream – Real-Time E-Commerce Analytics <i>Kafka, Spark, TimescaleDB, FastAPI, React, Docker</i> GitHub	Mar. 2026
- Architected distributed platform with Apache Kafka (3 brokers) and Spark Structured Streaming processing 10K+ events/sec at <100ms latency with exactly-once delivery semantics and fault-tolerant consumer groups - Built FastAPI backend with automated anomaly detection and React dashboard with real-time WebSocket push; deployed via Docker Compose orchestrating 6 services; validated throughput under 3x peak load	
Biometric Attendance System <i>Python, Flask, dlib, PyTorch, PostgreSQL, Redis, Swift, SwiftUI, iBeacon</i>	Dec. 2025 – Present
- Designed multi-layer verification combining facial recognition (99%+ accuracy, dlib ResNet-34 CNN), iBeacon BLE proximity ($\pm 2m$), and liveness detection; replaced QR system after identifying credential-sharing vulnerability - Built native iOS app (Swift/SwiftUI/AVFoundation/CoreLocation) and Flask backend processing 128-dim face encodings; reduced attendance marking from 60s to under 5s at sub-2s response	
SPL – Arithmetic Expression Parser <i>Node.js, Express.js, D3.js</i> Live Demo GitHub	Oct. 2025
Built recursive descent parser with full PEMDAS precedence and interactive AST visualization via D3.js with expandable nodes, zoom, and pan; designed modular architecture decoupling lexer, parser, and renderer; deployed on Vercel	
GoWeather – Smart Weather Assistant <i>Go, Gin, MySQL, JavaScript, OpenWeatherMap API</i> GitHub	2025
Built production Go (Gin) backend with third-party API integration, relational MySQL schema, and responsive animated frontend; implemented concurrent session handling and database caching for performance	

TECHNICAL SKILLS

Languages: Python, Java, Go, JavaScript/TypeScript, C++, Swift, SQL
Backend & APIs: FastAPI, Flask, Spring Boot, Gin, Node.js, Express.js, Django, REST, gRPC, WebSockets, Microservices
Frontend: React, TypeScript, D3.js, SwiftUI, HTML/CSS, iOS (AVFoundation, CoreLocation, BLE/iBeacon)
Databases: PostgreSQL, MySQL, MongoDB, Redis, TimescaleDB
Cloud & DevOps: AWS (EC2, S3, Lambda, RDS), Docker, Docker Compose, GitHub Actions CI/CD, Git, Vercel
AI & Data: PyTorch, TensorFlow, LangChain, OpenCV, Apache Kafka, Spark, Claude Code, Cursor IEEE ICICV 2024, ICCES 2023